ECE382M-7 Final Project Proposal

Group Name: Group 13

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1. **Introduction**

   Modern system-on-chip designs often integrate intellectual property blocks that use different on-chip bus protocols. This project proposes a synthesizable Verilog RTL bridge that allows a **TileLink Uncached Lightweigh (TL-UL)** host to access an AXI4-Lite peripheral. The bridge will translate transactions, preserve response information, and handle independent channel backpressure.

2. **Design Plan**

   The implementation will support TL-UL operations are Get, PutFullData, and PutPartialData; the corresponding responses are AccessAckData for reads and AccessAck for writes. The RTL will contain request capture and decode logic, independent AXI read and write channel controllers, and response generation.

   For a write, it will track AW and W handshakes independently and issue exactly one TL response after the AXI B response. AXI error responses will assert the TL d_error field.

   Verification will cover reads, full and partial writes, delayed responses, independent AW/W acceptance, backpressure on both interfaces, reset, and AXI error responses. Randomized stalls will supplement these directed tests.

3. **Design Goals**

   a. Functional correctness: complete each supported read or write exactly once, return the correct TL response, and preserve data, byte enables, source, size, and error status.

   b. Protocol correctness: obey VALID/READY handshakes on both interfaces, including stalls on any AXI channel and either TL channel; retain payloads while a transfer is stalled.

   c. Physical implementation: deliver synthesizable RTL, a completed APR run, and reproducible area and timing reports.

4. **Citations/References**

https://opentitan.org/book/hw/ip/tlul/index.html?utm_source=chatgpt.com

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